

Marta Markowicz

martasm2@illinois.edu | [LinkedIn](#) | [Website](#) | +1 612-760-5645

EDUCATION

University of Illinois Urbana-Champaign

Ph.D. in Computer Science (Robotics and Motion Planning)

Expected May 2027

Urbana, IL

University of Minnesota Twin Cities

B.S. in Computer Science, GPA 3.9

2017-2021

Minneapolis, MN

TECHNICAL SKILLS

Robotics: Motion Planning, Multi-Robot Planning, Sampling-Based Planning, Autonomous Navigation, Controls, Model Predictive Control, Real-Time Systems

Machine Learning: Reinforcement Learning, Robot Learning, Deep Learning, Model-Based RL

Programming Languages: C++, Python, Java

Frameworks and Environments: ROS/ROS 2, OMPL, PyTorch, CUDA, Gazebo, CMake, Linux, Git

WORK EXPERIENCE

Ph.D. Researcher: Robotics & Motion Planning

2021-Present

University of Illinois — Advisor: Dr. Nancy Amato

Urbana, IL

- Leading a five-person research team developing motion planning algorithms for dynamic and multi-robot environments
- Developed SPITE, a dynamic roadmap for replanning in changing environments, achieving up to 60% faster updates and reducing preprocessing from hours to minutes [C.1]
- Developed hierarchical swept-volume methods for multi-robot conflict checking, reducing motion-planning time by up to 50% [R.1]
- Extended dynamic roadmaps with lazy validation to accelerate task and motion planning [W.2]
- Contributed to [Open-SPITE](#), an open-source C++ library integrating SPITE with OMPL and VAMP

Applied Scientist Intern

May-Aug 2025

Amazon Robotics

North Reading, MA

- Developed and optimized graph-based path planning algorithms for autonomous mobile robots transporting large carts through warehouse environments
- Reduced graph-search time by 50% by decreasing graph complexity and path post-processing time by 80% by pre-optimizing node connections
- Increased multi-agent throughput by 25% through these optimizations

Robotics Research Assistant

2017-2021

University of Minnesota — Advisor: Dr. Stephen J. Guy

Minneapolis, MN

- Developed a sampling-based planner in C++ using temporal safe intervals to achieve asymptotically time-optimal planning with 2x faster near-optimal convergence
- Reduced predicted path collisions by 70% versus a space-time RRT baseline in unknown dynamic environments using uncertainty-aware obstacle models, evaluated on real human trajectory data

Autonomy Software Design Intern

Jun-Sept 2020

Caterpillar

Brooklyn Park, MN

- Developed software for autonomous soil-compactor field testing, reducing on-site monitoring requirements by 40%
- Built a C++/Qt interface for real-time remote management of five autonomous machines

Software Development Intern

Jan-Mar 2019

Cybercom Poland

Lodz, Poland

- Integrated Arduino-based sensing hardware with AWS Lambda for real-time collection and monitoring of pipe-inspection data
- Automated inspection workflows, saving city technicians approximately 60 hours per month
- Developed AI chatbot to assist patients of medical provider

PROJECTS

Autonomous Vehicle Motion Planning

Spring 2024

Tools: ROS, Gazebo, Python, C++

- Designed and implemented Hybrid A* and Model Predictive Control (MPC) motion-planning components on a Polaris GEM e2 vehicle.

Real-Time Systems for Path Planning

Fall 2021

Tools: Autoware, C++

- Parallelized planning tasks with virtual gang scheduling, achieving an 8% improvement in execution time

Model-Based Reinforcement Learning for Efficient Exploration

Fall 2021

Tools: MuJoCo, PyTorch

- Implemented ME-TRPO, a model-based RL approach using an ensemble of learned dynamics models for policy optimization
- Evaluated extensions using expert trajectories, model-ensemble disagreement, and PPO-based policy optimization for continuous-control tasks

Solar Vehicle Project: Controls

2017-2019

Tools: FreeRTOS, C++, Altium

- Led a five-person team designing and fabricating a PCB for a steering-wheel control system
- Implemented real-time control software in FreeRTOS with CAN communication and diagnosed hardware/software failures during international competition

PUBLICATIONS

C=CONFERENCE, P=PREPRINT, R=UNDER REVIEW, W=WORKSHOP

- [R.1] M Markowicz, J Motes, M Morales, N Amato, "MR SPITE: Accelerating Multi-Robot Motion Planning via Hierarchical Swept-Volume Approximations," *arXiv preprint; under review for ICRA 2027*.
- [P.1] Y Arad, S Ashur, M Markowicz, J Motes, M Morales, N Amato, "Serialized Red-Green-Gray: Quicker Heuristic Validation of Edges in Dynamic Roadmap Graphs," *arXiv preprint, 2026*.
- [W.1] Y Arad, M Markowicz, J Motes, M Morales, N Amato, "Open SPITE," *RoboARCH Workshop @ ICRA 2026*. Code
- [W.2] M Markowicz, S Ashur, J Motes, N Amato, "Semi-Lazy Rearrangement Solver for Accelerating Task Planning Using SPITE Dynamic Roadmaps," *Construction Workshop @ ICRA 2025*.
- [C.1] S Ashur, M Markowicz, M Lusardi, J Motes, M Morales, S Har-Peled, N Amato, "SPITE: Simple Polyhedral Intersection Techniques for modified Environments," *WAFR 2024*.

LEADERSHIP AND AWARDS

Mentoring Undergraduate Researchers

2021-Present

- Supervise undergraduate research in motion planning, including simulation evaluation of Multi-Robot SPITE and contributions to research publications
- Guided students in implementing and benchmarking motion-planning algorithms and contributing to an open-source planning library

Mentoring CS Research Scholars

2022-2026

- Mentored up to 50 undergraduate women per semester on research opportunities, coursework, and career development

Andrew and Shana Laursen Fellowship

2021

- Awarded to top graduate students in recognition of academic excellence and promise in advanced research

Hopper-Dean Scholarship honoring Dr. Vipin Kumar

2020

- Honored for academic excellence and outstanding contributions to undergraduate research



Website